



Department of Computer Science and Engineering
Premier University

CSE 457: Machine Learning

CT-03 Assignment

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Remarks

AI's Full Response

AI-Based Approach for Fraud Detection

To solve this fraud detection problem, the following AI-based approach is recommended:

1. Data Preprocessing

- Handle missing values and outliers
- Normalize numerical features like transaction amount
- Encode categorical features (location, device type)

2. Handling Imbalanced Data

- Use techniques like SMOTE (Synthetic Minority Over-sampling Technique)
- Alternatively use undersampling or class-weighted models

3. Feature Engineering

- Extract time-based features (hour, day, weekend)
- Create behavioral features from transaction history
- Detect unusual spending patterns

4. Model Selection

- Start with Logistic Regression (baseline)

- Use tree-based models like Random Forest and Gradient Boosting (XGBoost, LightGBM)
- These models perform well on tabular and imbalanced data

5. Evaluation Metrics

- Accuracy is not suitable due to imbalance
- Use Precision, Recall, F1-score
- ROC-AUC and PR-AUC are important

6. Model Tuning

- Use cross-validation
- Hyperparameter tuning with Grid Search or Random Search

7. Deployment

- Build API for real-time prediction
- Monitor model performance and retrain periodically

8. End-to-End Pipeline

- Data collection → preprocessing → feature engineering → training → evaluation → deployment → monitoring

The most suitable models are Gradient Boosting models (like XGBoost or LightGBM) due to their strong performance on structured data and ability to handle imbalance.

Critical Evaluation

The AI's answer is quite good and detailed. It properly handles the imbalance problem using XGBoost and SMOTE, which is very important. It also discusses feature engineering and real-time systems effectively.

However, it has some weaknesses, such as the lack of specific accuracy numbers from other datasets and limited discussion about Bangladesh Bank regulations. Some parts of the explanation feel overly general.

LightGBM or XGBoost with `scale_pos_weight` and SHAP is likely the most suitable model choice because it is both efficient and interpretable.

System Architecture Design

